

Resolution Semantics for Partial Algebras

Intrinsic Finite-Complement Separation and Observational Completion, with a Lean-Checked Many-Sorted Encoding Bridge

Adrian Puha

17 August 2026

Abstract

We study one-sorted binary partial algebras through a normalized free compatible completion. Applications defined in the base algebra reduce to embedded base elements, while applications undefined there remain structured finite-tree Answers. The generated Answer algebra has the expected universal property and is the expression algebra modulo equality after resolution. On this established free-completion substrate, the main object of study is the class of compatible total extensions that fix the base carrier pointwise and have finite complement over it.

Every distinct pair of generated Answers is separated by such an extension. At each budget, separation by this class is equivalent to separation by the canonical finite-tag models

$$D.\text{Carrier} \sqcup \text{Fin}(n) \sqcup \{\star\}.$$

The number of fresh tags is bounded by the combined constructor size; when the base carrier is equipped with a finite code this gives finite separating targets with an explicit total-state bound. Finite-tag agreement induces a separated filtration whose Cauchy completion supports canonical operations and preserves exactly the universal equations in the language with named base-carrier constants. When the base carrier is finitely coded, the completion is proper exactly when some application in the base algebra is undefined. Without a finiteness assumption, a context fixing the base pointwise yields an explicit Cauchy orbit outside the generated image, strictly nontrivial finite stages, and unbounded separation ranks. Natural and integer arithmetic instantiate this criterion while retaining division by zero as a structured Answer outside the base image rather than an ordinary number.

To locate these results precisely within established partial-to-total methods, we also give a Lean-checked many-sorted development of Diaconescu’s encoding. It treats arbitrary finite arities and separate total and partial operation families, implements Γ , α , β , and γ , and checks the satisfaction, adjunction, semantic-consequence, and fixed-signature initiality results used here, including the operations-only existence step for the formalized quasi-existence theories. Its singleton-sort, no-*TF*, binary-*PF* instance is proved canonically equivalent to the original binary Resolution implementation, including the generated and truth carriers and all encoded operations. This bridge records provenance and formal compatibility; it is an explicit compatibility link between the two implementations. Lean 4.33.0 checks both layers.

Contents

1	Introduction	3
2	Generated Answers as a Free Compatible Completion	5
2.1	Expressions and Answers	5
2.2	Universal property	6
2.3	Quotient presentation	7
3	Intrinsic Finite-Complement Separation	7
3.1	Pair-dependent separator	8
4	The Observational Filtration and Completion	10
4.1	Relative universal property	13
4.2	Contrast with prefix-depth completion	13
5	Exact Equational Conservativity	14
6	Arithmetic Examples	14
6.1	Natural numbers	15
6.2	Integers	15
7	A Lean-Checked Bridge to Diaconescu’s Encoding	16
7.1	The encoded models, β , and α	16
7.2	The canonical model γ and its lift	17
7.3	The binary one-sorted specialization	17
7.4	The concrete adjunction, initiality, and semantic consequence	18
8	Relation to Prior Work	19
8.1	Free completion and partial algebras	19
8.2	Ground terms, recognizability, and operation trees	19
8.3	Tree metrics and completion	20
8.4	Finite complements and profinite comparison	20
8.5	Totalized division	20
8.6	Contribution and scope	21
9	Formalization and Trust Boundary	21
10	Limitations and Open Questions	23

11 Conclusion

23

1 Introduction

Free completion and partial-to-total encoding provide the established starting point for this paper. Climent Vidal and Cosme Llópez prove that the inclusion of total algebras into partial algebras has a left adjoint and develop the functorial Schmidt construction [28]. Diaconescu gives the many-sorted Horn-style encoding of partial algebras as total algebras [18]; it is the closest encoding framework to the normalized construction used here. Related completion questions also occur for partial combinatory algebras and term-rewriting models [7, 8, 23]. Accordingly, neither the existence of a free compatible extension nor the partial-to-total encoding itself is claimed as new.

Against that background, the paper studies a more specific binary question: separation by compatible total extensions that preserve an arbitrary partial base pointwise while only their complement over that base is required to be finite, and the observational completion induced by all such finite budgets. The main separation, properness, and completion results are one-sorted and binary. To make their dependence on established encoding theory precise, [section 7](#) develops a many-sorted Lean comparison with Diaconescu’s construction and proves that its singleton-sort, no-*TF*, binary-*PF* instance agrees canonically with the original binary Resolution implementation.

Let D be a one-sorted binary partial algebra with carrier A , operation symbols Σ , and evaluator

$$\text{eval}_D : \Sigma \longrightarrow A \longrightarrow A \longrightarrow \text{Option}(A).$$

The value $\text{some}(c)$ records that the application is defined in D and equals c ; none records that it is undefined in D . We call A *finitely coded* when it admits an injection into $\text{Fin}(m)$ for some m . Classically this is equivalent to finiteness, but retaining the explicit code is mathematically useful: the parameter m enters the quantitative bounds $m + n + 1$ for stage- n targets and $m + |x| + |y| + 1$ for pair-dependent separating targets.

Classical tree-automata results imply ordinary residual finiteness for the finite-base, finite-signature ground-equational presentation discussed below [14, 20, 27]; that qualitative special case is not claimed as new either.

Our starting presentation is operationally simple. Resolve an expression bottom-up. If a node has two embedded base operands and the evaluator of D returns a value, replace the node by that embedded base value. Otherwise retain the node, its operation symbol, and its resolved children. The resulting finite trees are called *Answers*. This gives a normalized representative for a free compatible total extension.

On top of the established free-completion substrate, the paper asks two narrower questions.

- (Q1) Can equality of generated Answers be characterized by total models that do not themselves contain Answer trees, while preserving every application defined in the base algebra?
- (Q2) Can those finite observations support a completion on which the operations remain canonical and all constant-bearing universal equations are unchanged?

The answer to both is yes. The following two statements summarize the principal binary Resolution results.

Theorem 1.1 (Structured free completion and finite separation). *For every one-sorted binary partial algebra D :*

- (a) *the generated Answer algebra $R(D)$ is initial among compatible total algebras equipped with a map from the base carrier;*
- (b) *$R(D)$ is represented by the quotient of the absolutely free expression algebra by the kernel of the resolution evaluator;*
- (c) *every distinct $x, y \in R(D)$ is separated by a compatible total extension whose complement over A has at most $|x| + |y| + 1$ points; equivalently, a canonical finite-tag model needs at most $|x| + |y|$ fresh tags and one overflow state. Consequently, when A is finitely coded, $R(D)$ is residually finite.*

Theorem 1.2 (Observational completion and equation preservation). *Finite-tag agreement defines a separated descending filtration on $R(D)$. Its Cauchy completion $\widehat{R(D)}$ is complete, contains an injective stage-dense image of $R(D)$, and carries the unique jointly nonexpansive extensions of the primitive operations. Relative to complete, separated filtered compatible total extensions whose base embedding is injective, the completed algebra has the expected universal extension property. If the base carrier is finitely coded, the canonical embedding is surjective exactly when the evaluator of D is total; equivalently, the completion is proper exactly when some application in D is undefined. Without assuming finite coding of the base carrier, suppose some application in D is undefined and there are $u \in \Sigma$ and $e \in A$ with $\text{eval}_D(u, a, e) = \text{some}(a)$ for every $a \in A$. Then every finite observation stage is strictly coarser than equality, an explicit factorial orbit is Cauchy with no generated limit, its separation ranks are unbounded, and the completion embedding is not surjective. Moreover, for every pair of constant-bearing algebraic terms s, t ,*

$$\widehat{R(D)} \models s = t \iff R(D) \models s = t.$$

The next theorem records the supporting many-sorted formalization that locates the binary construction within Diaconescu's established encoding.

Theorem 1.3 (Many-sorted encoding bridge). *Let $\Sigma = (S, TF, PF)$ be a many-sorted partial-algebra signature whose operations have arbitrary finite arities. In the concrete operations-only Lean presentation developed here:*

- (a) *the Horn clauses Γ define a reduct β and the translation α satisfies $M \models \alpha(\rho)$ if and only if $\beta(M) \models \rho$;*
- (b) *every partial Σ -algebra D has a canonical encoded model $\gamma(D)$ satisfying Γ , and $\beta(\gamma(D))$ is canonically equivalent to D as a partial algebra;*
- (c) *every partial homomorphism $D \rightarrow \beta(M)$ has a unique encoded lift $\gamma(D) \rightarrow M$ agreeing with it under that equivalence;*
- (d) *the induced maps on homomorphisms respect identities and composition, the lift is an explicit hom-set equivalence natural in both variables, and its unit and counit satisfy both triangle identities;*
- (e) *at a fixed signature and universe bound, γ sends initial partial models to initial encoded models, while the β reduct of an initial model of $\Gamma \cup \alpha(E)$ is an initial partial model of E ; for every theory E in the concrete syntax of quasi-existence equations, a semantic conditional term model supplies the required initial encoded model and hence an initial partial model; moreover, for arbitrary sets E, E' of closed formulas,*

$$E \models_{PA} E' \iff \alpha(E) \cup \Gamma \models_{ALG} \alpha(E').$$

- (f) *when the signature has one sort, no designated-total symbols, and only binary partial symbols, the generated carrier and the canonical γ model are equivalent to the pre-existing binary Resolution implementation, with operations, existence equality, and the truth constant preserved in both directions.*

The first two clauses of [theorem 1.1](#) place the generated construction within established free-completion theory. The more specific ingredients are the intrinsic finite-complement observer class over a pointwise-preserved base, its pairwise separator, and the observational structure it induces. The linear estimate is retained as a general construction bound rather than a state-complexity advance. Over a finitely coded base it also gives an explicit, base-faithful bound on the entire separating target; ordinary residual finiteness itself is classical in the finite-signature subcase. The finite-base properness criterion explains exactly when the Cauchy construction adds genuinely new points over a finitely coded base carrier; the base-fixing-context criterion shows that properness is not confined to finite bases. The two hypotheses are incomparable: the one-point algebra is finitely coded and partial but has no base-fixing context, whereas natural arithmetic has a base-fixing context but no finite code for its carrier. Natural and integer arithmetic are concrete corollaries. [Theorem 1.3](#) supplies supporting formal provenance by connecting the binary construction to Diaconescu’s established encoding.

2 Generated Answers as a Free Compatible Completion

2.1 Expressions and Answers

Finite input expressions and raw Answers are defined by separate grammars:

$$e ::= \text{val}(a) \mid \text{app}(f, e_1, e_2),$$

$$r ::= \text{old}(a) \mid \text{susp}(f, r_1, r_2).$$

The separation is intentional. Expressions are input syntax. Answers are normal forms produced by the partial evaluator together with the suspension rule. The constructor $\text{old}(a)$ is retained as notation for the embedded element $\eta_0(a)$; in the prose these are called base elements.

Definition 2.1 (One-step resolution). For $f \in \Sigma$ and raw Answers x, y , let

$$\text{liftOp}_D(f, x, y) = \begin{cases} \text{old}(c), & x = \text{old}(a), y = \text{old}(b), \text{eval}_D(f, a, b) = \text{some}(c), \\ \text{susp}(f, x, y), & \text{otherwise.} \end{cases}$$

The resolution function Res_D is the structural recursion determined by

$$\text{Res}_D(\text{val}(a)) = \text{old}(a),$$

$$\text{Res}_D(\text{app}(f, e_1, e_2)) = \text{liftOp}_D(f, \text{Res}_D(e_1), \text{Res}_D(e_2)).$$

The generated carrier is the image of this evaluator:

$$R(D) = \{r \mid \exists e, \text{Res}_D(e) = r\}.$$

The canonical base embedding is $\eta_0(a) = \text{old}(a)$. Generated operations are

$$f_{R(D)}(x, y) = \text{liftOp}_D(f, x, y).$$

Generation is closed under these operations because expression witnesses can be joined by one application node.

Proposition 2.2 (Conservativity on defined applications). *If $\text{eval}_D(f, a, b) = \text{some}(c)$, then*

$$f_{R(D)}(\eta_0(a), \eta_0(b)) = \eta_0(c).$$

The map η_0 is injective.

The proposition is immediate from [definition 2.1](#). Its role is not depth but control: no calculation defined in D is reinterpreted.

2.2 Universal property

A *compatible total algebra over D* consists of a total Σ -algebra T , a map $i : A \rightarrow T$, and the preservation law

$$\text{eval}_D(f, a, b) = \text{some}(c) \implies f_T(i(a), i(b)) = i(c).$$

The map i is not required to be injective. This is the target notion needed for the ordinary free-completion universal property.

Every raw Answer has an interpretation in T :

$$\Phi_T(\text{old}(a)) = i(a), \quad \Phi_T(\text{susp}(f, x, y)) = f_T(\Phi_T(x), \Phi_T(y)).$$

The preservation law gives

$$\Phi_T(\text{liftOp}_D(f, x, y)) = f_T(\Phi_T(x), \Phi_T(y)).$$

The only non-structural case is the branch defined in D , where the equation is exactly the compatibility condition.

Theorem 2.3 (Free compatible completion). *For every compatible total algebra T over D , there is a unique homomorphism*

$$\Phi_T : R(D) \longrightarrow T$$

such that $\Phi_T \circ \eta_0 = i$.

Proof. Existence is the fold above. It preserves generated operations by the one-step equation. For uniqueness, choose an expression e generating a given Answer. Induction on e forces every compatible homomorphism to agree with the fold on $\text{Res}_D(e)$. Since every generated Answer has such a witness, the two maps agree everywhere. $\square$

The Lean-facing theorem is `ResolutionSemantics.freeCompletionUniversal`. It identifies the generated construction as a concrete normalized presentation of the free-completion universal property.

2.3 Quotient presentation

Define an equivalence on input expressions by

$$e \sim_D e' \iff \text{Res}_D(e) = \text{Res}_D(e').$$

Then resolution descends to a map

$$\overline{\text{Res}}_D : \text{Expr}(D)/\sim_D \longrightarrow R(D).$$

Proposition 2.4 (Kernel quotient presentation). *The map $\overline{\text{Res}}_D$ is injective and surjective. Thus generated Answers are a normalized presentation of the expression algebra modulo the resolution kernel.*

The two inverse laws are formalized by `ResolutionSemantics.expressionKernelQuotientInjective` and `ResolutionSemantics.expressionKernelQuotientSurjective`.

Remark 2.5 (Relation to the Schmidt construction). The Schmidt construction and the left adjoint free-completion functor are direct prior art [28]. Our formal theorem proves the corresponding universal property for the concrete one-sorted binary presentation used here. A fully explicit categorical isomorphism between the Lean object and every formulation of the many-sorted Schmidt construction is not needed for the results below. Thus $R(D)$ is positioned as a normalized free compatible completion.

3 Intrinsic Finite-Complement Separation

The free-completion theorem alone does not show that distinctions between suspended Answers are visible in independently specified models. The primitive observer condition should not depend on a particular sum encoding.

Let T be a compatible total extension with injective base embedding $i : A \hookrightarrow T$. Define its genuine outside part by

$$\text{Out}_D(T) = \{z \in T \mid \forall a \in A, z \neq i(a)\}.$$

Definition 3.1 (Intrinsic finite-complement observer). An observer has budget n when there is an injection

$$\text{Out}_D(T) \hookrightarrow \text{Fin}(n+1).$$

Thus only the complement over the pointwise-preserved base is finite; the base carrier itself may be infinite. The budget is an upper bound, so no ordering, chosen tags, or distinguished overflow point is part of this definition.

Every compatible extension has the carrier decomposition

$$T \cong A \sqcup \text{Out}_D(T).$$

For the concrete observer used in the construction below, the outside part is canonically $\text{Fin}(n) \sqcup \{\star\}$.

Definition 3.2 (Finite-tag model). For $n \in \mathbb{N}$, a finite-tag model over D has carrier

$$A \sqcup \text{Fin}(n) \sqcup \{\star\}$$

and total operations preserving every application defined in D :

$$\text{eval}_D(f, a, b) = \text{some}(c) \implies f_T(\text{old}(a), \text{old}(b)) = \text{old}(c).$$

No further equations are imposed.

The formal representation theorem proves that intrinsic separation at budget n is equivalent to finite-tag separation at the same budget. In the forward direction, an intrinsic complement code yields a base-fixing injection into the canonical carrier. Unused tag states are allowed and are sent to overflow when the operations are transported. This is an injection presentation, not a claimed bijection with all canonical states.

Write $x \equiv_n y$ when every finite-tag model with n tags interprets x and y equally. Full finite observational agreement means

$$x \equiv_{\text{fin}} y \iff \forall n, x \equiv_n y.$$

3.1 Pair-dependent separator

Let $x \neq y$ be generated Answers. Form a finite list $S(x, y)$ containing all subterms of the two raw trees. Repetition is harmless; first-occurrence indices provide a finite set of tags. Define an encoding on selected nodes:

$$q(\text{old}(a)) = \text{old}(a), \quad q(s) = \text{tag}(\text{index}(s))$$

for selected suspended nodes s . The encoding is injective on the selected closure.

The model operation first preserves an application defined in D . Otherwise it attempts to decode both inputs, constructs one suspended node, and re-encodes that node if it belongs to the selected closure; all unrelated cases go to $\star$. There is no branch conflict. A generated normal Answer cannot contain a suspended node with two base children whose application is defined in D , because resolution would already have reduced it.

Induction on selected subterms gives

$$\Phi_T(s) = q(s).$$

In particular, $\Phi_T(x) = q(x)$ and $\Phi_T(y) = q(y)$, so injectivity of q on $S(x, y)$ separates the roots.

Theorem 3.3 (Intrinsic finite-complement separation). *For distinct $x, y \in R(D)$ there exist n and a compatible total extension T whose outside part injects into $\text{Fin}(n + 1)$ and such that*

$$\Phi_T(x) \neq \Phi_T(y).$$

One may choose

$$n = |x| + |y|,$$

where $|-|$ is raw constructor count. Consequently,

$$x \equiv_{\text{fin}} y \iff x = y.$$

The intrinsic public Lean declarations are `ResolutionSemantics.qualitativeFiniteComplementSeparating_theorem`, `ResolutionSemantics.intrinsicSeparationIffFiniteTag`, and `ResolutionSemantics.intrinsicExtensionHasCanonicalPresentation`. The canonical finite-tag consequences remain exposed as `ResolutionSemantics.finiteExternalSeparation`, `ResolutionSemantics.finiteObservationComplete`, and `ResolutionSemantics.finiteSeparationBound`.

Remark 3.4 (Interpretation of the result). The base carrier A may be infinite. The theorem is therefore not finite residuality of the whole carrier. It says intrinsically that a chosen pair requires only finitely many *new* semantic states beyond A ; the finite-tag sum is a derived normal form. The displayed linear estimate is a baseline construction bound, not an optimality theorem. The mathematical content is the explicit model class and separator; no additional “full abstraction” terminology is needed.

More precisely, let $\mathcal{C}_D$ be the class of compatible total extensions that fix A pointwise and have finite complement over A . The theorem says that $R(D)$ is residually $\mathcal{C}_D$: every unequal pair is separated by its canonical homomorphism into some member of $\mathcal{C}_D$. This is an exact classification of the statement as a relative residual property, not a claim that the terminology or general residual paradigm is new.

Corollary 3.5 (Finite-base residual finiteness). *Suppose the base carrier admits an injective code into $\text{Fin}(m)$. Then $R(D)$ is residually finite in the ordinary homomorphic sense. More precisely, distinct x, y are separated by a finite compatible algebra with an injective code into*

$$\text{Fin}(m + |x| + |y| + 1).$$

The formal statements are `ResolutionSemantics.ResidualComparison.finiteBaseCompatibleSeparationBound` and `ResolutionSemantics.ResidualComparison.finiteBaseGeneratedResiduallyFinite`. The target produced by the proof still embeds the base carrier injectively, so the formal result is slightly stronger than ordinary residual finiteness, whose separating homomorphisms need not be faithful on the chosen generators.

Remark 3.6 (Finite-signature ground-equation comparison). Assume in addition that the operation signature is finite. Regard each $a \in A$ as a constant and let E_D consist of the finitely many ground equations

$$f(a, b) = c \quad \text{for every } \text{eval}_D(f, a, b) = \text{some}(c).$$

Every expression is congruent modulo E_D to a term presenting its resolved Answer. Conversely, every equation in E_D is respected by resolution. Thus equality after resolution is exactly the congruence generated by E_D .

Kozen proved that the classes of a finitely generated term congruence are recognizable tree languages [20]. For one such class, its finite-index syntactic congruence contains the defining congruence: congruent terms remain congruent in every context and therefore have the same membership in that class. The syntactic quotient separates the chosen class from every different one, so ordinary residual finiteness in this finite-base, finite-signature subcase follows from that classical result. Foundational regular-tree systems, finite ground rewriting, standard tree-automata treatments, and many-sorted recognizability theorems provide the broader context [9, 14, 16, 27]. That qualitative consequence is therefore classical. The explicit construction above retains the base carrier injectively, gives the stated whole-target bound, and continues to arbitrary base carriers by bounding only the complement; those are the features used in the remainder of the paper. The Lean corollary itself does not assume that the operation signature is finite.

Proposition 3.7 (Closure of the base image). *Let $\iota_D : A \hookrightarrow R(D)$ be the canonical base embedding. Its image is closed under every total operation of $R(D)$ if and only if the evaluator of D is total.*

Indeed, an application defined in D reduces to an embedded base value. If $\text{eval}_D(f, a, b) = \text{none}$, however, then

$$f^{R(D)}(\iota_D(a), \iota_D(b)) = \text{susp}(f, \text{old}(a), \text{old}(b)),$$

which is not in the base image. Lean checks both the pointwise escape statement `ResolutionSemantics.ResidualComparison.undefinedApplicationEscapesOldImage` and the equivalence `ResolutionSemantics.ResidualComparison.generatedOldClosedIffEvaluatorTotal`.

The selected-subterm encoding together with an overflow state has familiar tree-automata antecedents [14]. Its use here is constrained by compatibility with a pointwise-preserved partial base. The resulting targets are generally not finite: A may be infinite, while only the added tags and $\star$ are counted. A further computational analogy is the separating-words problem. Sublinear worst-case bounds were established by Goralčík and Koubek, Robson, and Chase [12, 19, 25]. Along a unary comb, a Resolution observer likewise iterates one transition map from a fixed starting state. This analogy confirms that our linear estimate is a baseline, but it does not transfer word bounds automatically: arbitrary Answer trees branch, and every observer must extend the pointwise-preserved partial algebra. The resulting relative separation complexity is open. Determining whether this exact observer class already has a standard name remains part of our specialist literature audit.

4 The Observational Filtration and Completion

Each $\equiv_n$ is an equivalence relation. The formal model family yields a descending filtration: agreement at a larger observational budget implies agreement at every smaller budget. By [theorem 3.3](#),

$$\bigcap_{n \in \mathbb{N}} \equiv_n = \{(x, x) \mid x \in R(D)\}.$$

Thus the filtration is separated.

This filtration is not the classical prefix-depth filtration on finite trees. The explicit contrast is established in [proposition 4.5](#) after the completion construction.

A symbolic non-Archimedean distance can be read from the least separating stage. Equal points have distance zero; a distinct pair receives the level of its least finite separator. The strong triangle law follows from transitivity of every stage relation. This is standard filtered-space machinery, analogous to established metric treatments of trees [1, 15].

A sequence (x_k) is Cauchy when, for every stage n , its tail is pairwise $\equiv_n$ -equivalent. Two Cauchy sequences are asymptotically equivalent when they eventually agree at every fixed stage. Let

$$\widehat{R(D)} = \{\text{Cauchy sequences in } R(D)\} / \sim.$$

The constant-sequence map is injective by separatedness and stage-dense by the usual late-term argument. A diagonal construction proves completeness.

Theorem 4.1 (Finite-base properness criterion). *Suppose the base carrier A admits an injection into $\text{Fin}(m)$ for some m . Then the canonical embedding*

$$R(D) \longrightarrow \widehat{R(D)}$$

is surjective if and only if the evaluator of D is total. Equivalently, this embedding is not surjective if and only if there exist $f \in \Sigma$ and $a, z \in A$ with $\text{eval}_D(f, a, z) = \text{none}$.

Proof. A stage- n observer has at most $m + n + 1$ states: the finitely coded base carrier, n fresh tags, and overflow. If $\text{eval}_D(f, a, z)$ is undefined, let c_k be the left comb obtained by iterating this application. Along the comb family every observer acts by the orbit of one self-map. Hence $(c_k!)_{k \in \mathbb{N}}$ is Cauchy: sufficiently late factorial indices have difference divisible by every possible orbit period.

For non-convergence, fix a proposed finite Answer x . Choose the preserving finite-tag observer constructed from the finite subterm closure of x together with a comb prefix longer than x . It encodes x and the prefix by distinct non-overflow states. The next comb transition matches no selected normalized node, so it enters overflow; no selected transition has overflow as an encoded child, making overflow absorbing along the rest of the comb. Thus this single finite stage separates x from every sufficiently late factorial comb. No generated Answer is a limit.

Conversely, if the evaluator of D is total, structural induction shows that every generated Answer lies in the base image. Cauchy agreement at a fixed separating stage then makes every Cauchy sequence eventually constant, so the generated space is complete. Completeness is equivalent to surjectivity of the canonical completion embedding. The checked public statements are `ResolutionSemantics.FiniteBaseCompletion.completeIffTotal` and `ResolutionSemantics.FiniteBaseCompletion.embeddingNotSurjectiveIffExistsUndefined`. $\square$

The one-point always-undefined algebra is the smallest corollary and still provides an explicit represented completion point through `ResolutionSemantics.OnePointCompletion.addsPoint`. The theorem does not by itself decide infinite-base examples. The following independent criterion does.

Theorem 4.2 (Properness from a base-fixing context). *Let D be any binary partial algebra; no finiteness assumption is made on its base carrier. Suppose*

$$\text{eval}_D(g, a, b) = \text{none} \quad \text{and} \quad \text{eval}_D(u, x, e) = \text{some}(x) \quad (x \in A)$$

for some $g, u \in \Sigma$ and $a, b, e \in A$. Define

$$c_0 = \text{susp}(g, \text{old}(a), \text{old}(b)), \quad c_{k+1} = \text{susp}(u, c_k, \text{old}(e)).$$

Then:

- (a) $c_1 \neq c_2$ but $c_1 \equiv_0 c_2$;
- (b) *for every m , the distinct pair $c_{(m+2)!}, c_{(m+3)!}$ agrees at stage m , and its least finite-tag separation rank is greater than m ;*
- (c) $(c_k!)_{k \in \mathbb{N}}$ *is Cauchy and has no generated limit.*

Consequently the generated filtered space is not complete, the finite-tag separation ranks are unbounded, and the canonical completion embedding is not surjective.

Proof. Let T be a stage- m preserving observer and let $F(s) = u_T(s, \text{old}(e))$. Preservation gives $F(\text{old}(x)) = \text{old}(x)$ for every $x \in A$. For this orbit, collapse the entire base carrier to one settled state while retaining the m fresh tags and overflow. The collapse commutes with F and has $m + 2$ states. Equality in the collapsed orbit lifts to the original orbit: complement states were retained injectively, while an orbit that reaches a base value $\text{old}(x)$ stays at that exact value because $F(\text{old}(x)) = \text{old}(x)$. Finite-orbit periodicity therefore implies that sufficiently late indices whose difference is divisible by $(m + 2)!$ have equal

T -values. Applying this simultaneously to factorial indices proves that (c_{kl}) is Cauchy and gives the stage- m agreement in clause (b). Constructor count proves the pair distinct. Since a distinct pair agrees at stage m , the exact rank characterization makes its least separating budget greater than m .

At stage zero the complement consists only of overflow. If F sends overflow to a base value, the next application fixes that value; if it sends overflow to overflow, the same table entry fixes it. Thus $F^2 = F$, proving $c_1 \equiv_0 c_2$, while constructor count again proves $c_1 \neq c_2$.

For non-convergence, fix a proposed finite Answer x . The standard candidate-tailored preserving observer encodes x together with a repeated $u(-, e)$ prefix longer than x . The next iterate is absent from the selected finite table and therefore enters overflow; overflow is absorbing along the remaining tail. The candidate remains encoded by a non-overflow state, so one finite stage separates x from every sufficiently late factorial term. Hence no generated Answer is a limit. The checked public statements are `ResolutionSemantics.OldFixingContextCompletion.factorialSequenceCauchy`, `ResolutionSemantics.OldFixingContextCompletion.factorialSequenceNoLimit`, `ResolutionSemantics.OldFixingContextCompletion.separationRankGreaterThanOrEqualToBudget`, and `ResolutionSemantics.OldFixingContextCompletion.embeddingNotSurjective`. $\square$

Natural and integer arithmetic both satisfy the hypotheses with undefined seed $0/0$ and base-fixing context $x \mapsto x + 0$. Thus [theorem 4.2](#) proves that both arithmetic completion embeddings are non-surjective and that neither filtration has a finite stage equal to equality. The direct public corollaries include `ResolutionSemantics.NatDivision.oldFixingCriterion`, `ResolutionSemantics.NatDivision.separationRanksUnbounded`, `ResolutionSemantics.IntDivision.completionEmbeddingNotSurjective`, and `ResolutionSemantics.IntDivision.separationRanksUnbounded`.

Proposition 4.3 (Incomparability of the properness hypotheses). *Neither properness hypothesis subsumes the other.*

- (i) *The one-point always-undefined algebra has a one-state code, an application undefined in the base algebra, and a proper completion, but it admits no base-fixing-context witness.*
- (ii) *Natural arithmetic has the base-fixing witness $x \mapsto x + 0$ and a proper completion, but its base carrier admits no finite code.*

Proof. For the one-point algebra, any base-fixing witness would require its sole operation to satisfy $\text{eval}_D(u, \star, \star) = \text{some}(\star)$, contradicting the definition $\text{eval}_D \equiv \text{none}$. Its one-state code, undefined operation, and proper completion are explicit.

For natural arithmetic, the witness is the undefined seed $0/0$ together with the context $x \mapsto x + 0$. If a finite code of bound m existed, applying the bounded pigeonhole theorem to the $m + 1$ inputs $0, \dots, m$ would give distinct i, j with equal codes, contradicting injectivity. The checked statements are `ResolutionSemantics.PropernessCriteria.onePointProperWithoutOldFixing` and `ResolutionSemantics.PropernessCriteria.natProperWithoutFiniteCoding`. $\square$

Proposition 4.4 (Completed operations). *Every generated operation $f_{R(D)}$ is jointly stage-compatible and therefore extends pointwise to a unique jointly nonexpansive operation*

$$\widehat{f} : \widehat{R(D)} \times \widehat{R(D)} \longrightarrow \widehat{R(D)}.$$

For embedded points,

$$\widehat{f}(\eta x, \eta y) = \eta(f_{R(D)}(x, y)).$$

The completed carrier and operations form a compatible total extension of D with injective base embedding. Its completeness is exposed by `ResolutionSemantics.completionComplete`; the compatible algebra is `ResolutionSemantics.completedAlgebra`.

4.1 Relative universal property

The formalization defines a category-like class of targets consisting of:

- a compatible total extension of D with injective base embedding;
- a separated descending filtration;
- nonexpansive generated interpretation;
- jointly nonexpansive primitive operations;
- completeness of the filtered carrier.

For every such target T , generated interpretation extends uniquely from the stage-dense subalgebra, and binary-extension uniqueness shows that the extension preserves all primitive operations. Hence the completion is initial relative to this specified target class. The exact Lean package is `Resolution.External.FilteredTotalAlg.completedResolutionFilteredAlgebra_initial`.

The adjectives “initial,” “reflective,” and “idempotent” are mathematically correct only with these hypotheses; without them they would express a different categorical statement.

4.2 Contrast with prefix-depth completion

Ultrametric and algebraic treatments of infinite trees use observations by prefix depth [1, 15]. Write $x \equiv_n^d y$ when two Answers agree at every level of depth strictly less than n . This is the filtration whose completion gives the classical infinite-tree space. It differs genuinely from finite-tag observation.

Proposition 4.5. *Let D be a binary partial algebra with $\text{eval}_D(f, a, z) = \text{none}$ for some f, a, z . Define the combs by $c_0 = a$ and $c_{k+1} = f(c_k, z)$. Then the sequence $(\text{Res}_D c_k)_{k \in \mathbb{N}}$ is Cauchy for the filtration $\equiv_n^d$ but not Cauchy for the filtration $\equiv_n$.*

The formal statement is `Resolution.Probe.depthCauchy_not_observationalCauchy`. Each comb suspends at every level, so $\text{Res}_D c_i$ and $\text{Res}_D c_j$ agree at every depth strictly less than n whenever $i, j \geq n$: under the tree metric the sequence converges to the infinite comb. For the finite-tag filtration one preserving model with a *single* fresh tag suffices to break Cauchy convergence. Send an undefined pair of base elements to the tag, the tag to $\star$, and $\star$ back to the tag; the fold along the combs then alternates between the tag and $\star$. Thus adjacent combs are separated at stage 1 for every k , however deep the disagreement lies (`Resolution.Probe.comb_separatesAt_one`). A fixed observational budget therefore sees through unbounded depth, which prefix truncation cannot do. Consequently the identity on $R(D)$ does not carry $\equiv_n^d$ -Cauchy sequences to $\equiv_n$ -Cauchy ones, and no uniformly continuous isomorphism of the two completions commutes with the canonical embeddings. The statement about the two filtrations is formalized; the consequence for the completions is the standard argument for uniformly continuous maps.

The same comb family shows that the general construction bound need not be tight. Since $|\text{Res}_D c_k| = 2k + 1$, the bound in [theorem 3.3](#) permits $4k + 4$ tags to separate c_k from c_{k+1} , whereas the preceding observer uses one.

Open Question 4.1 (Optimal relative separation complexity). For distinct Answers let their separation rank be the least n for which $x \not\equiv_n y$. What is the optimal worst-case growth of this rank in $|x| + |y|$, and how does it depend on finiteness of the signature and of the base carrier? [Theorem 3.3](#) gives a linear upper bound, while the comb family shows that this bound can be far from tight.

5 Exact Equational Conservativity

Let algebraic terms contain variables, named base-carrier constants, and binary operation symbols. They can be evaluated in $R(D)$ and in $\widehat{R(D)}$. Embedding commutes with term evaluation by induction. Moreover, every term operation preserves each stage relation because the primitive operations do.

Fix a completed assignment ρ and a stage n . Stage-density allows us to choose, independently for every variable, a generated assignment ρ_n with

$$\rho(v) \equiv_n \eta(\rho_n(v)).$$

Stage compatibility then gives

$$\llbracket s \rrbracket_\rho \equiv_n \eta(\llbracket s \rrbracket_{\rho_n})$$

and similarly for t .

Theorem 5.1 (Exact equation preservation). *For every constant-bearing equation $s = t$,*

$$\left(\forall \rho : \text{Var} \rightarrow R(D), \llbracket s \rrbracket_\rho = \llbracket t \rrbracket_\rho \right)$$

if and only if

$$\left(\forall \hat{\rho} : \text{Var} \rightarrow \widehat{R(D)}, \llbracket s \rrbracket_{\hat{\rho}} = \llbracket t \rrbracket_{\hat{\rho}} \right).$$

Proof. For the forward implication, approximate an arbitrary completed assignment at every stage. The generated equation identifies the two embedded approximants, so the completed term values agree at every stage. Separatedness yields exact equality. The reverse implication follows by evaluating on embedded generated assignments and using injectivity of the embedding. $\square$

The public theorem is `ResolutionSemantics.equationConservative`.

Remark 5.2 (Conditional equations). The theorem concerns universal equations. Arbitrary quasi-equations do not follow from density alone. The formal development transfers conditional laws only under explicit stage-robust or dense-premise hypotheses. No unrestricted quasi-equational conservativity claim is made.

6 Arithmetic Examples

The general construction does not depend on ring or field axioms. We include natural-number and integer instances because they make the behavior at a zero denominator concrete.

6.1 Natural numbers

Take operations $+$, $\cdot$, and $/$ on $\mathbb{N}$, with division undefined exactly when the denominator is zero. For $b \neq 0$,

$$\text{divide}(a, b) = \text{old}(a/b).$$

At zero,

$$\text{divide}(a, 0) = \text{susp}(\text{div}, \text{old}(a), \text{old}(0)).$$

Therefore

$$a/0 = b/0 \implies a = b,$$

and no $a/0$ is an embedded base natural. These properties remain true after the completion embedding. They are exposed by `ResolutionSemantics.NatDivision.singularFamilyInjective` and `ResolutionSemantics.NatDivision.singularFamilyDisjoint`.

The observational completion is nontrivial in a stronger sense. By [theorem 4.2](#), repeatedly applying the context $x \mapsto x + 0$, which fixes the base carrier pointwise, produces distinct finite Answers that no fixed observation stage can fully separate. The factorially sampled family is Cauchy without a generated limit, so the completion contains an explicit point outside the generated image, and the explicit pair at budget m has separation rank greater than m . This uses the base-fixing-context criterion rather than the finite-base theorem.

The contrast with prefix depth is sharper in this instance. For the iterated singular divisions $(\dots((a/0)/0)\dots/0)$, no fresh tags are needed: sending each singular division to the successor of its numerator gives a preserving model over $\mathbb{N} \sqcup \{\star\}$ that separates every pair of distinct combs at stage 0 (`Resolution.Probe.NatProbe.natComb_pairwise_separated_at_zero`). This does not make the full natural-arithmetic filtration discrete. That observer uses new base naturals generated along the iterated-division table; the family $c_0 = 0/0$, $c_{k+1} = c_k + 0$ cannot do so because preservation forces $a + 0 = a$ on every base natural. In fact c_1 and c_2 agree in every stage-zero observer, and [theorem 4.2](#) supplies an indistinguishable distinct pair at every finite stage.

6.2 Integers

The integer instance additionally includes subtraction. Again, every nonzero division agrees with the evaluator of the base algebra, while

$$a/0 = \text{susp}(\text{div}, \text{old}(a), \text{old}(0)).$$

The completed singular family is injective, and completed $0/0$ is not any embedded integer. The paper-facing results are `ResolutionSemantics.IntDivision.singularFamilyInjective` and `ResolutionSemantics.IntDivision.zeroDivZeroNotOld`.

Integer addition also has zero as a right identity on the base carrier. Hence the same base-fixing-context criterion proves that every finite integer-arithmetic stage is strictly coarser than equality, its separation ranks are unbounded, and its completion embedding is not surjective. These corollaries are `ResolutionSemantics.IntDivision.everyFiniteStageNotEquality`, `ResolutionSemantics.IntDivision.factorialOldFixingCauchy`, and `ResolutionSemantics.IntDivision.completionEmbeddingNotSurjective`.

The construction does not assign an ordinary numerical value to $0/0$. It gives $0/0$ a structured Answer in a conservative extension.

This distinction matters when comparing with totalized arithmetic. In an involutive meadow one usually sets $0^{-1} = 0$; in a common meadow exceptional terms propagate one designated error; wheels adopt a different algebraic extension with infinity- and error-like behavior [3, 6, 11]. Resolution chooses provenance retention rather than a fixed base value or one undifferentiated error.

7 A Lean-Checked Bridge to Diaconescu's Encoding

The preceding sections develop the binary separation-completion results. We now record their exact connection to the total-algebra encoding of [18]. This section verifies the inherited foundation and the interface used by Resolution. Diaconescu works with a many-sorted signature $\Sigma = (S, TF, PF)$, where symbols in TF are total and symbols in PF are partial. The Lean structure assigns each symbol a natural-number arity, a sort at every finite argument position, and a result sort. Neither the set of sorts nor either symbol family is required to be finite.

7.1 The encoded models, β , and α

An encoded total model M consists of a data carrier M_s for each $s \in S$, an auxiliary truth carrier M_b , total interpretations of all TF and PF symbols on the data carriers, operations

$$e_s : M_s \times M_s \longrightarrow M_b, \quad \text{true} \in M_b,$$

and the Horn conditions Γ . The implemented clauses say that total symbols preserve and reflect definedness, partial symbols reflect definedness, and

$$x \ e_s \ y = \text{true} \implies x \ e_s \ x = \text{true} \quad \text{and} \quad x = y.$$

Thus the carrier of the reduct is

$$\beta(M)_s = \{x \in M_s \mid x \ e_s \ x = \text{true}\}.$$

Designated-total operations restrict to total operations on these subtypes; a PF symbol is defined precisely when its totalized result is marked as defined. Encoded homomorphisms induce partial-algebra homomorphisms between the corresponding reducts.

The formula syntax is indexed by a list of bound-variable sorts. Typed de Bruijn variables make quantifier extension explicit. It is generated by existence-equality atoms, conjunction, negation, and one-variable universal quantification; iterating the last constructor represents a finite sorted variable block. The translation sends

$$t \stackrel{e}{=} t' \longmapsto (t \ e_s \ t' = \text{true}),$$

commutes with the Boolean constructors, and guards a total universal quantifier by definedness of the newly bound variable.

The two term lemmas corresponding to Lemmas 3.3 and 3.4 of [18] are `Resolution.Diaconescu.ManySorted.term_evalPartial_sound` and `Resolution.Diaconescu.ManySorted.term_evalPartial_complete`. Structural induction on formulas, with the guard handling the quantifier case, then gives `Resolution.Diaconescu.ManySorted.alpha_satisfaction_condition`. The theorem is stated for assignments and hence also yields the closed-sentence instance in [theorem 1.3\(a\)](#).

7.2 The canonical model γ and its lift

For a partial algebra D , raw values at each sort are either embedded base values or formal applications. The normalizer treats the two symbol families differently:

- a TF application whose arguments are base values always reduces to the embedded total result;
- a PF application whose arguments are base values reduces exactly when the source partial evaluator returns a value;
- every remaining application is retained as a sorted formal suspension with its normalized arguments.

The generated subtype consists of raw values having a term witness. Its interpreter into any compatible total target exists and is unique. This is the many-sorted finite-arity free-completion argument used for the data sorts of $\gamma(D)$.

The auxiliary carrier of $\gamma(D)$ has one true element and a failure value for each sorted pair that is not the same diagonal base value. Existence equality returns true exactly on that base diagonal. Lean checks that this model satisfies all clauses of Γ in `Resolution.Diaconescu.ManySorted.gamma_satisfiesGamma`. The base embedding is bijective onto the defined part, and it respects both the total and partial evaluators; this is packaged by `Resolution.Diaconescu.ManySorted.betaGammaPartialAlgEquiv`. Finally, `Resolution.Diaconescu.ManySorted.gamma_persistent_liberality` constructs the lift of every $D \rightarrow \beta(M)$ and proves uniqueness from the free interpreter on data and the homomorphism laws on the truth carrier.

7.3 The binary one-sorted specialization

The many-sorted normal-form development above is implemented autonomously from the earlier binary core. The module `ResolutionDiaconescuBinarySpecialization.lean` supplies the formal comparison rather than relying on their structural resemblance. It sends a binary signature to a universe-polymorphic singleton data sort, an empty family of designated-total symbols, and the original operation family as partial symbols of arity two. Forgetting and restoring the unique sort gives the partial-algebra equivalence `Resolution.Diaconescu.BinarySpecialization.toFromPartialAlgEquiv`, and the three binary Horn clauses are shown equivalent to the specialized many-sorted Γ record.

At the free-completion level, the interpreters in the two directions are constructed from the respective universal properties. Uniqueness proves that their composites are identities, yielding `Resolution.Diaconescu.BinarySpecialization.generatedEquiv`. The equivalence preserves base elements and generated binary operations. It also transports the base-diagonal predicate, failure witnesses, and the auxiliary truth carrier. These components are assembled in `Resolution.Diaconescu.BinarySpecialization.gammaEquiv`, which formally identifies the original binary $\gamma(D)$ with the binary one-sorted reduct of the many-sorted $\gamma(D)$. Its forward and inverse maps commute with the operation symbols, existence equality, and the distinguished truth value. Thus the general layer and the binary layer use different Lean types but agree canonically on their shared mathematical fragment.

7.4 The concrete adjunction, initiality, and semantic consequence

The continuation module makes the categorical content explicit while keeping the formal dependency surface small. It proves the identity and composition laws for β , packages persistent liberality as the checked bijection

$$\text{Hom}_{\text{ALG}_\Gamma}(\gamma(D), M) \cong \text{Hom}_{PA}(D, \beta(M)),$$

and defines the induced action of γ on homomorphisms. The latter respects identities and composition. The hom-set equivalence is natural in both variables; its unit $D \rightarrow \beta\gamma(D)$ and counit $\gamma\beta(M) \rightarrow M$ are explicit and satisfy both triangle identities. These statements are exposed by `Resolution.Diaconescu.ManySorted.betaHom_comp`, `Resolution.Diaconescu.ManySorted.gammaBetaHomEquiv`, and `Resolution.Diaconescu.ManySorted.gammaHom_comp`, together with `Resolution.Diaconescu.ManySorted.gammaRestrict_natural_target`, `Resolution.Diaconescu.ManySorted.gammaRestrict_natural_source`, `Resolution.Diaconescu.ManySorted.gammaBetaCounit_natural`, and `Resolution.Diaconescu.ManySorted.gammaBeta_triangle_gamma`.

The left-adjoint content gives a direct initial-semantics consequence. For a fixed signature and a fixed carrier universe closed under γ , define an initial partial model of a theory E by the usual unique-map property among models of E , and define initiality on the encoded side among models of Γ and $\alpha(E)$. The theorem `Resolution.Diaconescu.ManySorted.gamma_preserves_initiality` proves that γ sends the former initial model to the latter. In the direction used by Diaconescu’s Corollary 4.2, `Resolution.Diaconescu.ManySorted.beta_of_initial_encoded_is_initial` proves that the β reduct of an initial encoded model is an initial partial model. This isolates the categorical transfer in the published argument.

The remaining existence step is supplied in two separate modules. First, `Resolution.Conditional.termModel_initial` treats an arbitrary many-sorted conditional-equational theory. Sorted ground terms are quotiented by semantic equality in every model of the theory. Lean proves that the resulting term algebra satisfies the theory and that evaluation induces the unique homomorphism from it to every other model.

Second, the syntax formalized for a quasi-existence equation consists of a finite sorted variable context, a finite list of existence-equality premises, and one existence-equality conclusion. Its conditional translation adds a definedness guard for every context variable and translates each atom using e_s and true . The equivalence between satisfaction of this conditional clause and satisfaction in the β reduct is `Resolution.Diaconescu.InitialExistence.translateQuasiEquation_sat_iff_partial`. The five families of Γ clauses are included in the same conditional theory. Applying the term-model construction to $\Gamma \cup \alpha(E)$ gives the initial encoded model in `Resolution.Diaconescu.InitialExistence.initialEncodedModel_is_initial`; the adjunction transfer then yields `Resolution.Diaconescu.InitialExistence.quasiTheory_has_initial_partial_model`. Thus the fixed-signature, same-universe, operations-only initial-model conclusion used in Corollary 4.2 is checked end to end, providing a concrete formalization of the established existence argument.

Because Lean represents $\beta(\gamma(D))$ as canonically equivalent to D rather than definitionally equal to it, reflection of consequence requires an explicit transport result. The theorem `Resolution.Diaconescu.ManySorted.Formula.sat_iff_of_equiv` proves invariance of satisfaction under partial-algebra equivalence, including the negation and sorted-quantifier cases. Combining this with the α - β satisfaction condition and the canonical γ model gives `Resolution.Diaconescu.ManySorted.semantic_consequence_equivalence`, the concrete Lean form of Corollary 4.3 of [18].

The data carrier is exposed as the many-sorted Resolution normal-form type. Theorems `Resolution.Diaconescu.ManySorted.gammaHom_generatedOld`, `Resolution.Diaconescu.ManySorted.gammaHom_generatedTotal`, and `Resolution.Diaconescu.ManySorted.gammaHom_generatedPartial` show that the functorial action preserves base values and the two generated operation constructors. This is the explicit interface between the established encoding and the Resolution normal-form layer used here; the specialization equivalence above connects it formally to the binary `GeneratedAns` carrier used in [sections 2 to 6](#).

There are two deliberate scope boundaries. The Lean theorem uses canonical partial-algebra equivalence rather than identifying $\beta(\gamma(D))$ and D by definitional equality. It does not mechanize the surrounding institution/comorphism infrastructure, arbitrary signature morphisms, the full institution-level form of Corollary 4.2, or the proof-theoretic Birkhoff completeness result of Section 5 of [18]. The categorical fixed-signature transfer and the quasi-existence initial-model construction used inside Corollary 4.2 are checked, but their packaging over arbitrary signature morphisms is not. These boundaries are why the result is described here as a many-sorted encoding bridge and not as a replacement for the broader theory of [18].

8 Relation to Prior Work

8.1 Free completion and partial algebras

The closest structural prior art is the free completion of partial algebras and the Schmidt construction [28]. Theorem 2.3 should be read as a Lean-checked concrete presentation of that general idea for the one-sorted binary setting. Other completion and encoding results already establish partial-to-total extension [7, 8, 18, 23]. In particular, [section 7](#) follows Diaconescu’s choices of Γ , α , β , and γ and checks them in the repository’s concrete typed syntax. Its value here is the verified interface between that established encoding and Resolution normal forms.

8.2 Ground terms, recognizability, and operation trees

Separating operation syntax from interpretation is standard in term algebras, free monads, and algebraic-effects semantics [24]. The specific normalization discipline used here reduces a node immediately when its application is defined in D and retains it with already resolved children when the application is undefined in D .

For finite bases and finite signatures this normalization is also a finite ground-equational presentation, as detailed in [remark 3.6](#). Recognizable tree languages, syntactic congruences, and finite tree automata are therefore direct prior art for the ordinary residual-finiteness consequence [9, 14, 16, 20, 27]. The pair-dependent separator likewise uses the standard finite closure of selected subterms plus a sink-like state. The distinguishing constraint in this paper is not that combinatorial pattern but its compatibility with a possibly infinite base that is fixed pointwise while only the complement is budgeted.

8.3 Tree metrics and completion

Ultrametric and algebraic treatments of infinite trees long predate this work [1, 15]. Our completion is distinguished by its source of finite observations: agreement in all compatible finite-tag models, rather than prefix depth. The Cauchy-sequence quotient and continuous extension arguments are standard once the separated filtration is available. The distinction itself is a result of the construction, not merely a bibliographic comparison: [proposition 4.5](#) exhibits a comb sequence that is prefix-depth Cauchy but fails the finite-tag Cauchy condition already at stage 1. The arithmetic strengthening appears in [section 6](#), and the resulting state-complexity problem is isolated as [Open Question 4.1](#).

8.4 Finite complements and profinite comparison

Finite-complement extensions are classical in semigroup theory under the name of finite Rees index. Cain and Maltcev’s survey explicitly records that residual finiteness passes both to finite-Rees-index subsemigroups and to their finite-complement extensions, citing the theorem of Ruškuc and Thomas [10, 26]. Thus residual finiteness is not a gap in that literature. Miller, O’Reilly, Quick, and Ruškuc study residual, weak, strong, and complete separability conditions for semigroups in the finite-quotient tradition [21].

These results are important adjacent prior art, but neither citation by itself identifies the observer class used here. Our base is a pointwise-fixed partial subalgebra, while a separator is a compatible total extension whose complement over that base is finite; the separating codomain need not itself be finite when the base is infinite. Moreover, ordinary Rees index starts with a closed subsemigroup. By [proposition 3.7](#), the base image in the generated Answer algebra is closed under the total operations exactly in the already total case. Hence finite Rees index is not literally the inclusion used by a genuinely partial Resolution algebra. The remaining specialist question is whether separation by finite-complement extensions of a pointwise-fixed *partial* base has already been studied under a generalized Rees-index, relative residual-finiteness, or separability formulation.

There is likewise a mature theory of profinite completions and natural extensions of residually finite algebras, as well as categorical formulations through profinite monads [13, 17]. Recent work on profinite trees gives a particularly close comparison on the tree side [22]. Those constructions are organized by finite quotients or finite recognizing algebras. Our stages instead allow a possibly infinite preserved base and bound only the genuinely new complement. The finite-base properness theorem above is therefore not presented as a theorem about ordinary profinite completion. The documented comparison identifies close analogues, but no exact equivalence or prior statement of the criterion in the cited sources.

8.5 Totalized division

Meadows, common meadows, wheels, non-involutive meadows, partial meadows, and fracpair constructions provide mature alternatives with strong equational and initial-algebra results [2–6, 11]. Resolution is not proposed as uniformly superior. Its design goal is different: preserve calculations defined in D while retaining the provenance of applications undefined in D .

8.6 Contribution and scope

The established ingredients include free completion, operation-tree syntax, finite-signature ground-term recognizability, and standard Cauchy-completion machinery. Against that background, the specific contribution of this paper is concentrated in two connected parts:

1. an intrinsic relative observer class that fixes an arbitrary, possibly infinite partial base pointwise and budgets only its complement, together with a canonical finite-tag presentation, pairwise separation, and an explicit base-faithful construction bound; the central result is [theorem 3.3](#);
2. the induced observational completion, including an exact finite-base properness criterion, an independent arbitrary-base base-fixing-context criterion, their formal incomparability, the contrast with prefix-depth completion, and exact universal equational conservativity; in particular, [theorem 4.2](#) extends the non-surjectivity argument beyond finitely coded bases.

The audited Lean development checks these binary results, their natural- and integer-arithmetic instances, and the explicit unbounded-rank witnesses. The many-sorted development of [section 7](#) has a different role: it verifies the inherited encoding theory and proves that the binary implementation used by the two items above is its exact specialization. That bridge supplies supporting formal provenance rather than a third item in this list. Our literature search has not identified the exact relative separation–completion package in the cited sources; the claim here is about the verified package, not historical priority for each constituent.

9 Formalization and Trust Boundary

The paper-only repository exposes a deliberately small public interface:

```
import ResolutionSemanticsCompletion
import ResolutionIntrinsicFiniteComplementAPI
import ResolutionFiniteBasePropernessPublic
import ResolutionOldFixingContextPropernessPublic
import ResolutionPropernessCriteriaComparison
import ResolutionArithmeticPropernessPublic
import ResolutionDiaconescuManySorted
import ResolutionDiaconescuContinuation
import ResolutionDiaconescuAdjunction
import ResolutionDiaconescuBinarySpecialization
import ResolutionDiaconescuInitialExistence

#check ResolutionSemantics.freeCompletionUniversal
#check ResolutionSemantics.OutsideOld
#check ResolutionSemantics.oldOrOutsideEquiv
#check ResolutionSemantics.outsideEquiv
#check ResolutionSemantics.qualitativeFiniteComplementSeparating_theorem
#check ResolutionSemantics.intrinsicExtensionHasCanonicalPresentation
#check ResolutionSemantics.finiteExternalSeparation
#check ResolutionSemantics.finiteSeparationBound
#check ResolutionSemantics.completionComplete
#check Resolution.External.FilteredTotalAlg.
    completedResolutionFilteredAlgebra_initial
```

```

#check ResolutionSemantics.equationConservative
#check ResolutionSemantics.FiniteBaseCompletion.completeIffTotal
#check ResolutionSemantics.FiniteBaseCompletion.
  embeddingNotSurjectiveIffExistsUndefined
#check ResolutionSemantics.OldFixingContextCompletion.embeddingNotSurjective
#check ResolutionSemantics.OldFixingContextCompletion.
  separationRankGreaterThanBudget
#check ResolutionSemantics.PropernessCriteria.onePointProperWithoutOldFixing
#check ResolutionSemantics.PropernessCriteria.natProperWithoutFiniteCoding
#check ResolutionSemantics.NatDivision.separationRanksUnbounded
#check ResolutionSemantics.IntDivision.completionEmbeddingNotSurjective
#check ResolutionSemantics.IntDivision.separationRanksUnbounded
#check Resolution.Diaconescu.ManySorted.term_evalPartial_sound
#check Resolution.Diaconescu.ManySorted.term_evalPartial_complete
#check Resolution.Diaconescu.ManySorted.alpha_satisfaction_condition
#check Resolution.Diaconescu.ManySorted.gamma_satisfiesGamma
#check Resolution.Diaconescu.ManySorted.betaGammaPartialAlgEquiv
#check Resolution.Diaconescu.ManySorted.gamma_persistent_liberality
#check Resolution.Diaconescu.ManySorted.betaHom_comp
#check Resolution.Diaconescu.ManySorted.gammaBetaHomEquiv
#check Resolution.Diaconescu.ManySorted.gammaHom_comp
#check Resolution.Diaconescu.ManySorted.Formula.sat_iff_of_equiv
#check Resolution.Diaconescu.ManySorted.semantic_consequence_equivalence
#check Resolution.Diaconescu.ManySorted.gammaHom_generatedOld
#check Resolution.Diaconescu.ManySorted.gammaHom_generatedTotal
#check Resolution.Diaconescu.ManySorted.gammaHom_generatedPartial
#check Resolution.Diaconescu.ManySorted.gammaRestrict_natural_target
#check Resolution.Diaconescu.ManySorted.gammaRestrict_natural_source
#check Resolution.Diaconescu.ManySorted.gammaBetaCounit_natural
#check Resolution.Diaconescu.ManySorted.gammaBeta_triangle_beta
#check Resolution.Diaconescu.ManySorted.gammaBeta_triangle_gamma
#check Resolution.Diaconescu.ManySorted.gamma_preserves_initiality
#check Resolution.Diaconescu.ManySorted.beta_of_initial_encoded_is_initial
#check Resolution.Diaconescu.BinarySpecialization.toFromPartialAlgEquiv
#check Resolution.Diaconescu.BinarySpecialization.generatedEquiv
#check Resolution.Diaconescu.BinarySpecialization.gammaEquiv
#check Resolution.Conditional.termModel_initial
#check Resolution.Diaconescu.InitialExistence.
  translateQuasiEquation_sat_iff_partial
#check Resolution.Diaconescu.InitialExistence.initialEncodedModel_is_initial
#check Resolution.Diaconescu.InitialExistence.
  quasiTheory_has_initial_partial_model

```

The repository pins Lean 4.33.0. The command

```
bash scripts/verify.sh
```

compiles the checked-in proof dependency chain, checks every manuscript-linked declaration, rejects direct proof-hole commands, and requires each audited public theorem to depend only on `propext`, `Classical.choice`, and `Quot.sound`. The separate command `bash scripts/build-paper.sh -check-committed` rebuilds this PDF deterministically and verifies that the checked-in copy matches. Finite indexing and witness selection use ordinary classical choice in documented places. The result is relative to Lean’s underlying logic and implementation; no foundation-independent consistency claim is made.

The paper gives full mathematical arguments for the free-completion and finite separator constructions because those determine its positioning. Routine completion lemmas are summarized and linked to exact declarations in `THEOREM_MAP.md`.

10 Limitations and Open Questions

1. For finite bases and finite signatures, ordinary residual finiteness is already explained by classical ground-term recognizability. The precise relation of the arbitrary-base, pointwise-preserving finite-complement observer class to known relative residual-finiteness constructions is not resolved by the cited literature.
2. The linear tag bound in [theorem 3.3](#) is a baseline upper bound, not an optimality theorem; its relative state complexity is [Open Question 4.1](#).
3. Properness is classified exactly for finitely coded base carriers. The base-fixing-context hypothesis is sufficient but not necessary, and neither of the two stated hypotheses subsumes the other; a full arbitrary-carrier classification remains open.
4. The $\Gamma\text{-}\alpha\text{-}\beta\text{-}\gamma$ bridge is many-sorted and supports arbitrary finite arities and separate *TF/PF* families. Its singleton-sort, no-*TF*, binary-*PF* instance is formally identified with the original binary Resolution core, but the finite-complement separation and observational-completion results remain one-sorted and binary. Extending those results and comparing them directly with the full Schmidt construction remains open.
5. The bridge checks the concrete operations-only sentence and model constructions used here, including the fixed-signature adjunction laws and both initiality transfers. For the explicit quasi-existence syntax, it also constructs the initial translated conditional model and the resulting initial partial model. It does not formalize institution morphisms, arbitrary signature morphisms, the institution-level form of Diaconescu's Corollary 4.2, or Birkhoff proof-theoretic completeness.
6. The construction is semantic, not an efficiency claim. Retaining full provenance can produce large finite trees.
7. No ordinary field law is asserted globally on singular Answers unless it is separately proved in the generated algebra.
8. Constant-bearing universal equations transfer exactly; conditional equations require additional hypotheses.

11 Conclusion

Resolution semantics gives a normalized structured presentation of a free compatible completion of a partial algebra. Its free-completion character and basic conservativity follow from established ideas and the reduction rule. The central additional construction studied here is intrinsic finite-complement separation over the pointwise-preserved base fragment. It separates every generated pair while adding only finitely many states beyond the base; a derived finite-tag normal form supplies the explicit witnesses and induces a separated observational filtration. For finitely coded bases this implies ordinary residual finiteness with an explicit whole-target bound; the qualitative finite-signature consequence also follows from classical ground-term recognizability. For genuinely partial algebras the base image is not a total subalgebra, so the finite-complement condition is not literally ordinary Rees index. The general linear estimate is a baseline; its sharper relative complexity

is open. Over every finitely coded base carrier, the completion is proper exactly when an application in the base algebra is undefined. This finite-base hypothesis and the base-fixing-context hypothesis are incomparable, as witnessed by the one-point algebra and natural arithmetic respectively. Standard completion methods then produce a complete total algebra with canonical operations and exactly the same universal equations in the language with named base-carrier constants.

The base-fixing-context theorem explains the arithmetic examples uniformly: for both natural and integer arithmetic every finite stage is strictly coarser than equality, the separation ranks are unbounded, and the factorial repeated- $+0$ family supplies an explicit completion point outside the generated image. At the same time, $0/0$ remains a structured element outside the canonical numerical image; the construction does not replace ordinary arithmetic with a new numerical convention.

The many-sorted Lean development locates this binary framework precisely relative to Diaconescu’s total-algebra encoding. It checks the concrete TF/PF Horn theory, the α - β satisfaction condition, the canonical γ lift, the fixed-signature adjunction and initiality results, and the operations-only existence step for the formalized quasi-existence theories. Its singleton-sort, no- TF , binary- PF instance is canonically equivalent to the original binary implementation, including the generated and truth carriers and all encoded operations. These results establish formal provenance and compatibility while keeping the established encoding results credited to their source. The complete development is reproducible in Lean.

References

- [1] A. Arnold and M. Nivat. The metric space of infinite trees: Algebraic and topological properties. *Fundamenta Informaticae*, 3(4):445–475, 1980. doi: 10.3233/FI-1980-3405.
- [2] J. A. Bergstra and C. A. Middelburg. Division by zero in non-involutive meadows. *Journal of Applied Logic*, 13(1):1–12, 2015. doi: 10.1016/j.jal.2014.10.001.
- [3] J. A. Bergstra and A. Ponse. Division by zero in common meadows, 2014. URL <https://arxiv.org/abs/1406.6878>. Version 4, revised 2021.
- [4] J. A. Bergstra and A. Ponse. Fracpairs and fractions over a reduced commutative ring. *Indagationes Mathematicae*, 27(3):727–748, 2016. doi: 10.1016/j.indag.2016.01.007.
- [5] J. A. Bergstra and J. V. Tucker. The rational numbers as an abstract data type. *Journal of the ACM*, 54(2):7:1–7:25, 2007. doi: 10.1145/1219092.1219095.
- [6] J. A. Bergstra, Y. Hirshfeld, and J. V. Tucker. Meadows and the equational specification of division. *Theoretical Computer Science*, 410(12–13):1261–1271, 2009. doi: 10.1016/j.tcs.2008.12.015.
- [7] I. Bethke, J. W. Klop, and R. C. de Vrijer. Completing partial combinatory algebras with unique head-normal forms. In *Proceedings of the Eleventh Annual IEEE Symposium on Logic in Computer Science*, pages 448–454. IEEE Computer Society Press, 1996. doi: 10.1109/LICS.1996.561461.
- [8] I. Bethke, J. W. Klop, and R. C. de Vrijer. Extending partial combinatory algebras. *Mathematical Structures in Computer Science*, 9(4):483–505, 1999. doi: 10.1017/S0960129599002832.
- [9] W. S. Brainerd. Tree generating regular systems. *Information and Control*, 14(2):217–231, 1969. doi: 10.1016/S0019-9958(69)90065-5.
- [10] A. J. Cain and V. Maltcev. For a few elements more: A survey of finite Rees index, 2013. URL <https://arxiv.org/abs/1307.8259>.
- [11] J. Carlström. Wheels—on division by zero. *Mathematical Structures in Computer Science*, 14(1):143–184, 2004. doi: 10.1017/S0960129503004110.

- [12] Z. Chase. Separating words and trace reconstruction. In *Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing*, pages 21–31. Association for Computing Machinery, 2021. doi: 10.1145/3406325.3451118.
- [13] L.-T. Chen, J. Adámek, S. Milius, and H. Urbat. Profinite monads, profinite equations, and Reiterman’s theorem. In *Foundations of Software Science and Computation Structures*, volume 9634 of *Lecture Notes in Computer Science*, pages 531–547. Springer, 2016. doi: 10.1007/978-3-662-49630-5_31.
- [14] H. Comon, M. Dauchet, R. Gilleron, F. Jacquemard, D. Lugiez, S. Tison, and M. Tommasi. Tree automata techniques and applications, 2008. URL <https://www.irif.fr/users/tata/>. Electronic monograph.
- [15] B. Courcelle. Fundamental properties of infinite trees. *Theoretical Computer Science*, 25(2): 95–169, 1983. doi: 10.1016/0304-3975(83)90059-2.
- [16] M. Dauchet and S. Tison. The theory of ground rewrite systems is decidable. In *Proceedings of the Fifth Annual IEEE Symposium on Logic in Computer Science*, pages 242–248. IEEE Computer Society Press, 1990. doi: 10.1109/LICS.1990.113750.
- [17] B. A. Davey, M. J. Gouveia, M. Haviar, and H. A. Priestley. Natural extensions and profinite completions of algebras. *Algebra Universalis*, 66(3):205–241, 2011. doi: 10.1007/s00012-011-0155-y.
- [18] R. Diaconescu. An encoding of partial algebras as total algebras. *Information Processing Letters*, 109(23–24):1245–1251, 2009. doi: 10.1016/j.ipl.2009.09.008.
- [19] P. Goralčík and V. Koubek. On discerning words by automata. In *Automata, Languages and Programming: 13th International Colloquium, ICALP 1986*, volume 226 of *Lecture Notes in Computer Science*, pages 116–122. Springer, 1986. doi: 10.1007/3-540-16761-7_61.
- [20] D. Kozen. Complexity of finitely presented algebras. In *Proceedings of the Ninth Annual ACM Symposium on Theory of Computing*, pages 164–177. ACM, 1977. doi: 10.1145/800105.803406.
- [21] C. Miller, G. O’Reilly, M. Quick, and N. Ruškuc. On separability finiteness conditions in semigroups. *Journal of the Australian Mathematical Society*, 113(3):402–430, 2022. doi: 10.1017/S1446788721000124.
- [22] V. Moreau. Profinite trees, through Lawvere theories and the lambda-calculus, 2024. URL <https://arxiv.org/abs/2402.13086>. Version 2, revised 2026.
- [23] G. A. R. Nesin. Completing partial algebra models of term rewriting systems. Master of logic thesis, Institute for Logic, Language and Computation, University of Amsterdam, 2011.
- [24] G. D. Plotkin and J. Power. Algebraic operations and generic effects. *Applied Categorical Structures*, 11(1):69–94, 2003. doi: 10.1023/A:1023064908962.
- [25] J. M. Robson. Separating strings with small automata. *Information Processing Letters*, 30(4): 209–214, 1989. doi: 10.1016/0020-0190(89)90215-9.
- [26] N. Ruškuc and R. M. Thomas. Syntactic and Rees indices of subsemigroups. *Journal of Algebra*, 205(2):435–450, 1998. doi: 10.1006/jabr.1997.7392.
- [27] J. C. Vidal and E. C. Llópez. Congruence based proofs of the recognizability theorems for free many-sorted algebras. *Journal of Logic and Computation*, 30(2):561–633, 2020. doi: 10.1093/logcom/exz032.
- [28] J. C. Vidal and E. C. Llópez. Functoriality of the Schmidt construction. *Logic Journal of the IGPL*, 31(5):822–893, 2023. doi: 10.1093/jigpal/jzac048.